

SECTION AAnswer **all** questions.

1. Bromine water can be used to test for alkenes.

(a) (i) State the expected colour change for a positive test for alkenes. [1]

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(ii) Draw the structure of the product formed when propene reacts with bromine water. [1]

(b) Identify another reagent that can be used to test for the presence of alkenes. [1]

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2. Bonds in hydrocarbons are formed by the overlap of orbitals between each atom.

(a) Draw an *s*-orbital and a *p*-orbital in the space below. [1]

(b) Name the type of bond shown in the diagram below. [1]

